Chems Eddine Adel KEBAILY

Master's Student (M1) in Signal and Image Processing | AI Engineering | AI Agents & Generative AI

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Profile

Master's student in Signal and Image Processing at the Université du Littoral Côte d'Opale, moving into **AI engineering**: **AI agents**, **generative AI** and **prompt engineering**. Built an **AI agent for administrative document processing** (OCR, LLM, n8n), CK-AI, my portfolio's AI agent, and classification projects (phishing, **Vision Transformer**). Two internships at **Ooredoo Algeria** in web (Spring Boot, React, SQL/Oracle) and mobile (Flutter) development, following Agile practices. **Internship from 1 April 2027 (up to 5 months)**: AI agents, generative AI, ML/DL, vision, NLP.

Education

2026 – 2027 **Master's degree (M1) in Signal and Image Processing** (ongoing), Université du Littoral Côte d'Opale (ULCO), France

2022 – 2025 **Bachelor's degree in Computer Systems**, M'Hamed Bougara University of Boumerdès, Algeria

2022 **Baccalaureate in Technical Mathematics, Civil Engineering** (with honours), Ali Chekiri High School, Chéraga

Professional Experience

Mobile Development Intern — Ooredoo Algeria **Dec. 2024 – Jan. 2025**

- Cross-platform mobile apps with **Flutter** across the full **Agile** cycle; exposure to datacenter and **hybrid cloud** (GCP, on-premise).

Web Development Intern — Ooredoo Algeria **Dec. 2023 – Jan. 2024**

- Developed a web application for managing cash flows of point-of-sale (POS) terminals with **Spring Boot** and **React**.
- Analysed and processed data with **SQL** queries; maintained and optimised **Oracle** databases.
- Implemented functional enhancements, **unit tests**, technical documentation, code reviews and Git versioning.

Projects

AI Agent for Administrative Document Processing (French public services) **2026**

- **OpenCV** image preprocessing (denoising, **Sauvola** binarisation, deskewing), **OCR**, structured field extraction with an **LLM** (Claude API), completeness checks; **n8n** orchestration with **human-in-the-loop** validation, FastAPI, Docker.
- On 40 labelled documents: **97.6%** field accuracy, **100%** classification accuracy, OCR error reduced **90-fold** by preprocessing. Code

CK-AI — AI agent on my portfolio (Claude API, tool use) **2026**

- Bilingual agent built with the **Claude API** and **tool use**: **6 tools** (profile lookup, matching an internship offer to my skills, 4 actions on the page), up to 5 steps per question.
- **Guardrails** (size and rate limits, prompt-injection protection) and **serverless** deployment on Netlify. Demo

Phishing Detection with Machine Learning — Bachelor's Final Project (research) **2025**

- Comparison of **16 classification models** (**XGBoost**, LightGBM, random forest, neural networks…) on 100,077 URLs with **Python** and **scikit-learn**; best: XGBoost, **89.6%** accuracy, +10.6 points over hand-written rules. Report

Animal Recognition (90 species) — Computer Vision

- Multi-class image classification with a pre-trained **Vision Transformer (ViT)** and **transfer learning** in **PyTorch** (Hugging Face), GPU training on Kaggle. Code

Skills

Generative AI & agents	AI agents, prompt engineering, Claude API, structured outputs, n8n, FastAPI
Image processing	OpenCV, OCR (Tesseract), denoising, binarisation, evaluation (CER, F1)
Machine learning & vision	PyTorch, TensorFlow, scikit-learn, Vision Transformer, transfer learning, Kaggle, Google Colab
Programming	Python, C, C++, Java, JavaScript, PHP, SQL (Oracle), HTML/CSS
Development	Spring Boot, React, Node.js, Flutter, REST APIs
Tools & methods	Docker, PostgreSQL, Git/GitHub, Linux (Ubuntu), Windows Server, VS Code; Agile, code reviews, unit testing
Languages	Arabic (native), French (B2), English (B2)

Student Activities and Training

2024 – 2026 **DevUp Club, University of Boumerdès** — co-manager of the Visual Department (2025–2026), Development member (2024–2025)

Feb. 2025 **"AI to Fullstack" Bootcamp** — DevUp Club (grade: Very Good)